

문제 5

II

(i) ^① Since $\lambda^T \beta$ is estimable,

$$\exists r \text{ s.t. } X^T X r = \lambda \iff \lambda^T = r^T X^T X$$

$$\lambda^T \hat{\beta} = r^T X^T X \hat{\beta} = r^T X^T y : \text{linear}$$

$$\textcircled{2} E[r^T X^T y] = r^T X^T X \beta = \lambda^T \beta$$

so, $r^T X^T y$ and $\lambda^T \hat{\beta}$: LHE of $\lambda^T \beta$

③ Let $a^T y$ be any LHE of $\lambda^T \beta$.

$$E[a^T y] = a^T X \beta = \lambda^T \beta \quad \forall \beta$$

$$\Rightarrow a^T X = \lambda^T$$

$$0 \leq \text{var}(a^T y - r^T X^T y)$$

$$= \text{var}((a^T - r^T X^T) y)$$

$$= \sigma^2 (a^T - r^T X^T)(a - X r)$$

$$= \sigma^2 (a^T a - r^T X^T a - a^T X r + r^T X^T X r)$$

$$= \sigma^2 (a^T a - r^T \lambda - \lambda^T r + \lambda^T r)$$

$$= \text{var}(a^T y) - \text{var}(r^T X^T y)$$

$$\Rightarrow \text{var}(r^T X^T y) \leq \text{var}(a^T y)$$

$$\therefore X^T X^T Y = X^T \hat{\beta} \text{ is BLUE of } X^T \beta \quad \square$$

$$(ii) \text{ let } \beta = (\mu, \tau_1, \tau_2, \tau_3)^T,$$

$$L = \tau_1 - 3\tau_2 + 2\tau_3$$

$$H_0: L = 2 \text{ vs } H_1: L \neq 2$$

$$\lambda = (0, 1, -3, 2)^T, \quad L = \lambda^T \beta$$

$$y_{i.} = \sum_{j=1}^3 y_{ij}, \quad \bar{y}_{i.} = \frac{1}{3} y_{i.}$$

$$X^T X = \begin{pmatrix} 4 & 3 & 3 & 3 \\ 3 & 3 & 0 & 0 \\ 3 & 0 & 3 & 0 \\ 3 & 0 & 0 & 3 \end{pmatrix}, \quad X^T Y = \begin{pmatrix} y_{.1.} \\ y_{.2.} \\ y_{.3.} \\ y_{.4.} \end{pmatrix}$$

$$\begin{aligned} r &= \begin{pmatrix} 0 \\ 1/3 \\ -1 \\ 2/3 \end{pmatrix}, \quad X^T X r = \begin{pmatrix} 4 & 3 & 3 & 3 \\ 3 & 3 & 0 & 0 \\ 3 & 0 & 3 & 0 \\ 3 & 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} 0 \\ 1/3 \\ -1 \\ 2/3 \end{pmatrix} \\ &= \begin{pmatrix} 0 \\ 1 \\ -3 \\ 2 \end{pmatrix} = \lambda \end{aligned}$$

$\Rightarrow \lambda^T \beta$ is estimable by thm 12.2b.

The BLUE of $L = \lambda^T \beta$ is

$$\begin{aligned}\hat{L} &= r^T X^T Y \\ &= \frac{1}{3} y_{1.} - y_{2.} + \frac{2}{3} y_{3.} \\ &= \bar{y}_{1.} - 3\bar{y}_{2.} + 2\bar{y}_{3.}\end{aligned}$$

$$\text{var}(\hat{L}) = \sigma^2 r^T X^T X r = \sigma^2 r^T \lambda$$

$$\begin{aligned}r^T \lambda &= 0 \cdot 0 + \frac{1}{3} \cdot 1 + (-1)(-3) + \frac{2}{3} \cdot 2 \\ &= \frac{14}{3}\end{aligned}$$

$$\text{var}(\hat{L}) = \frac{14}{3} \sigma^2$$

$$\text{rank}(X) = 3, \quad n = 9 \Rightarrow n - \text{rank}(X) = 9 - 3 = 6$$

$$s^2 = \frac{\text{SSE}}{6} = \frac{\sum_{i=1}^3 \sum_{j=1}^3 (y_{ij} - \bar{y}_{i.})^2}{6}$$

$$\frac{s^2}{\sigma^2} \sim \chi^2(6) \text{ \& } \lambda^T \hat{\beta} \perp s^2 \text{ by thm 12.39.}$$

Under $H_0: L = 2$,

$$\begin{aligned}T &:= \frac{\hat{L} - 2}{s \sqrt{14/3}} = \frac{(\bar{y}_{1.} - 3\bar{y}_{2.} + 2\bar{y}_{3.} - 2)/\sigma}{s \sqrt{14/3} / \sigma} \\ &= \frac{N(0,1)}{\sqrt{\chi^2(6)/6}} \sim t(6)\end{aligned}$$

2.

(i)

$$x \leq Q_F(p),$$

$$F_t(y) = (1-t)F(y) + t\delta_x$$

$$= (1-t)F(y) + t$$

$$= p$$

$$\Rightarrow F(y) = \frac{p-t}{1-t} \Rightarrow y = F^{-1}\left(\frac{p-t}{1-t}\right)$$

$$= IF(x; T, F) = \frac{\partial F^{-1}\left(\frac{p-t}{1-t}\right)}{\partial t} \Big|_{t=0}$$

$$= \frac{p-1}{f(Q_F(p))}$$

$$x > Q_F(p), F_t(y) = (1-t)F(y) + t\delta_x$$

$$= (1-t)F(y)$$

$$= p$$

$$\Rightarrow F(y) = \frac{p}{1-t} \Rightarrow y = F^{-1}\left(\frac{p}{1-t}\right)$$

$$\Rightarrow IF(x; T, F) = \frac{\partial F^{-1}\left(\frac{p}{1-t}\right)}{\partial t} \Big|_{t=0}$$

$$= \frac{p}{f(Q_F(p))}$$

$$IF(x; T, F) = \frac{p - \mathbb{1}_{\{x \geq Q_F(p)\}}}{f(Q_F(p))}$$

$$\gamma^*(T; F) = \sup_x \left| \frac{p - \mathbb{1}_{\{x \geq Q_F(p)\}}}{f(Q_F(p))} \right|$$

$$= \frac{\max\{p, 1-p\}}{f(Q_F(p))}$$

$$\text{For } p > \frac{1}{2}, \quad \gamma^*(T, F) = \frac{p}{f(Q_F(p))}$$

SUPPOSE $T(F)$ is B -robust as $p \rightarrow 1$, then

$\exists p_0 \in (\frac{1}{2}, 1)$ & $M < \infty$ s.t. $\frac{p}{f(Q_F(p))} \leq M, p_0 < p < 1$

Since $Q_F(p) \in (Q_F(p_0), \infty)$ as $p_0 < p < 1$,

$$f(y) \geq \frac{p_0}{M} \Rightarrow y > Q_F(p_0)$$

$$\Rightarrow 1 = \int_{-\infty}^{\infty} f(y) dy \geq \int_{-\infty}^{\infty} \frac{p_0}{M} dy = \infty. \quad *$$

$$\Rightarrow \sup_{p_0 < p < 1} \gamma^*(T; F) = \infty$$

So $T(F) = Q_F(p)$ is not B -robust

as $p \rightarrow 1$



$$(ii) \quad U := F(X) \sim \text{Unif}(0,1)$$

$$\mathbb{E}_F[X h(F(X))] = \int_0^1 Q_F(u) h(u) du$$

$$\mathbb{E}_F[h(F(X))] = \int_0^1 h(u) du$$

$$\Rightarrow T(F) = \frac{\int_0^1 Q_F(u) h(u) du}{\int_0^1 h(u) du}$$

For fixed $u \in (0,1)$, $\xi_u := Q_F(u)$.

$$F_t := (1-t)F + t\delta_x$$

$$\text{If } x > \xi_u, \text{ then } Q_{F_t}(u) = Q_F\left(\frac{u}{1-t}\right)$$

$$\begin{aligned} \text{So, } IF(x; Q_F(u), F) &= \frac{d}{dt} Q_F\left(\frac{u}{1-t}\right) \Big|_{t=0} \\ &= \frac{u}{f(\xi_u)} \end{aligned}$$

$$\text{If } x < \xi_u, \text{ then } Q_{F_t}(u) = Q_F\left(\frac{u-t}{1-t}\right)$$

$$\begin{aligned} \text{So, } IF(x; Q_F(u), F) &= \frac{d}{dt} Q_F\left(\frac{u-t}{1-t}\right) \Big|_{t=0} \\ &= -\frac{1-u}{f(\xi_u)} \end{aligned}$$

$$\Rightarrow IF(x; Q_F(u), F) = \frac{u - 1 \mathbb{I}\{x \leq Q_F(u)\}}{f(Q_F(u))}$$

$$\Rightarrow IF(x; T, F) = \lim_{T \rightarrow 0} \frac{T(F_x) - T(F)}{T}$$

$$= \frac{1}{\int_0^1 h(u) du} \lim_{T \rightarrow 0} \int_0^1 h(u) \frac{Q_F(u) - Q_F(u)}{T} du$$

$$= \frac{1}{\int_0^1 h(u) du} \int_0^1 h(u) \lim_{T \rightarrow 0} \frac{Q_F(u) - Q_F(u)}{T} du$$

$$= \frac{1}{\int_0^1 h(u) du} \int_0^1 h(u) IF(x; Q_F(u), F) du$$

$$= \frac{1}{\int_0^1 h(u) du} \int_0^1 h(u) \frac{u - \mathbb{I}\{x \leq Q_F(u)\}}{f(Q_F(u))} du.$$

$$u = F(y), y = Q_F(u), du = f(y) dy$$

$$IF(x; T, F) = \frac{1}{\int_0^1 h(u) du} \int_{-\infty}^{\infty} h(F(y)) \mathbb{I}\{F(y) - \mathbb{I}\{x \leq y\}\} dy$$

$$= \frac{\int h(F(y)) \mathbb{I}\{F(y) - \mathbb{I}\{x \leq y\}\} dy}{\int h(F(y)) dF(y)}$$

$$\mathbb{E}[A_F(x)] = \mathbb{E}_F \left[\int_{-\infty}^x h(F(y)) dy \right]$$

$$= \int_{-\infty}^{\infty} h(F(y)) P(x \geq y) dy$$

$$= \int_{-\infty}^{\infty} h(F(y)) [1 - F(y)] dy$$

$$A_F(x) - \mathbb{E}_F[A_F(x)]$$

$$= \int_{-\infty}^x h(F(y)) dy - \int_{-\infty}^{\infty} h(F(y)) [1 - F(y)] dy$$

$$= \int_{-\infty}^{\infty} h(F(y)) [1_{\{y \leq x\}} - 1 + F(y)] dy$$

$$= \int_{-\infty}^{\infty} h(F(y)) [F(y) - 1_{\{x < y\}}] dy$$

$$= \int_{-\infty}^{\infty} h(F(y)) [F(y) - 1_{\{x \leq y\}}] dy$$

$$IF(x; T, F) = \frac{\int h(F(y)) [F(y) - 1_{\{x \leq y\}}] dy}{\int h(F(y)) dF(y)}$$

$$= \frac{A_F(x) - \mathbb{E}_F[A_F(x)]}{\mathbb{E}_F[h(F(X))]}$$

$$\mathbb{E}_F[h(F(X))]$$

□

3.

$$X = UDV^T, U^T U = I_p, V^T V = VV^T = I_p$$

$$\alpha := V^T \beta = (\alpha_1, \dots, \alpha_p)^T$$

$$\hat{\beta} = (X^T X)^{-1} X^T y = V D^{-1} U^T y$$

$$\begin{aligned} \text{var}(\hat{\beta}) &= \sigma^2 V D^{-1} U^T U D^{-1} V^T \\ &= \sigma^2 V D^{-2} V^T \end{aligned}$$

$$\text{tr}(\text{var}(\hat{\beta})) = \sigma^2 \sum_{i=1}^p \frac{1}{d_i^2}$$

$$\text{bias}(\hat{\beta}) = E(\hat{\beta}) - \beta = 0$$

$$\text{MSE}(\hat{\beta}) = \sigma^2 \sum_{i=1}^p \frac{1}{d_i^2}$$

$$\hat{\beta}^R = V(D^2 + \lambda I_p)^{-1} D U^T y$$

$$\begin{aligned} \text{var}(\hat{\beta}^R) &= (V D^2 V^T + \lambda I_p)^{-1} V D U^T U D V^T (V D^2 V^T + \lambda I_p)^{-1} \sigma^2 \\ &= (V D^2 V^T + \lambda I_p)^{-1} V D^2 V^T (V D^2 V^T + \lambda I_p)^{-1} \sigma^2 \\ &= V(D^2 + \lambda I_p)^{-1} D^2 (D^2 + \lambda I_p)^{-1} V^T \sigma^2 \\ &= V \left\{ \frac{d_i^2}{(d_i^2 + \lambda)^2} \right\} V^T \sigma^2 \\ \text{tr}(\text{var}(\hat{\beta}^R)) &= \sigma^2 \sum_{i=1}^p \frac{d_i^2}{(d_i^2 + \lambda)^2} \end{aligned}$$

$$\begin{aligned}\text{bias}(\hat{\beta}^R) &= V \left(\frac{d_i^2}{d_i^2 + \lambda} \right) U^T U D U^T \beta - \beta \\ &= \left(V \left(\frac{d_i^2}{d_i^2 + \lambda} \right) V^T - I \right) \beta \\ &= V \left(\frac{-\lambda}{d_i^2 + \lambda} \right) V^T \beta\end{aligned}$$

$$\begin{aligned}\text{bias}(\hat{\beta}^R)^T \text{bias}(\hat{\beta}^R) &= \beta^T V \left(\frac{\lambda^2}{(d_i^2 + \lambda)^2} \right) V^T \beta \\ &= \text{tr} \left(\beta^T V \left(\frac{\lambda^2}{(d_i^2 + \lambda)^2} \right) V^T \beta \right) \\ &= \sum_{i=1}^p \frac{\lambda^2}{(d_i^2 + \lambda)^2} \alpha_i(\beta)^2\end{aligned}$$

$$\text{MSE}(\hat{\beta}^R) = \sum_{i=1}^p \left(\frac{d_i^2}{(d_i^2 + \lambda)^2} \sigma^2 + \frac{\lambda^2}{(d_i^2 + \lambda)^2} \alpha_i(\beta)^2 \right)$$

$$\begin{aligned}\Delta(\lambda) &:= \text{MSE}(\hat{\beta}) - \text{MSE}(\hat{\beta}^R) \\ &= \sigma^2 \sum_{i=1}^p \left[\frac{1}{d_i^2} - \frac{d_i^2}{(d_i^2 + \lambda)^2} \right] - \sum_{i=1}^p \frac{\lambda^2}{(d_i^2 + \lambda)^2} \alpha_i^2 \\ &= \sigma^2 \sum_{i=1}^p \frac{d_i^4 + 2d_i^2\lambda + \lambda^2 - d_i^4}{d_i^2(d_i^2 + \lambda)^2} - \sum_{i=1}^p \frac{\lambda^2}{(d_i^2 + \lambda)^2} \alpha_i^2 \\ &= \sigma^2 \sum_{i=1}^p \frac{\lambda(2d_i^2 + \lambda)}{d_i^2(d_i^2 + \lambda)^2} - \sum_{i=1}^p \frac{\lambda^2}{(d_i^2 + \lambda)^2} \alpha_i^2\end{aligned}$$

$$\frac{\Delta(\lambda)}{\lambda} = \sigma^2 \sum_{i=1}^p \frac{2d_i^2 + \lambda}{d_i^2(d_i^2 + \lambda)^2} - \lambda \sum_{i=1}^p \frac{\alpha_i^2}{(d_i^2 + \lambda)^2}$$

$\forall \delta > 0,$

$$\lim_{\lambda \downarrow 0} \frac{A(\lambda)}{\lambda} = \sigma^2 \sum_{i=1}^p \frac{2d_i^2}{d_i^2 d_i^4} = 0$$

$$= 2\sigma^2 \sum_{i=1}^p \frac{1}{d_i^4} > 0$$

def

$$\Leftrightarrow \exists \delta > 0 \text{ s.t. } 0 < \lambda < \delta \Rightarrow \frac{A(\lambda)}{\lambda} > 0.$$

Since $\lambda > 0$, $A(\lambda) > 0$

Hence, for any sufficiently small $\lambda > 0$,

$$MSE(\hat{\beta}^R) < MSE(\hat{\beta}).$$

$$\exists \lambda > 0 \text{ s.t. } MSE(\hat{\beta}^R) \leq MSE(\hat{\beta})$$

□